Experiment 1: Chemically Induced Recruitment and SMO Ubiquitination

Purpose

Test whether bringing the E3 ubiquitin ligase MGRN1 to SMO, directly or through the MEGF8 intracellular domain, increases SMO ubiquitination and lowers SMO abundance. Recruitment is induced chemically with AP21967 (the DmrA/DmrC system) or without an inducer through ALFA-tag recognition by NbALFA. The constructs are expressed in NIH/3T3 cells.

Constructs

All ten constructs were ordered pre-cloned from GenScript in pcDNA3.1(+)-based mammalian expression vectors.

DmrA/DmrC recruitment system (induced with AP21967)

Construct	**Purpose**
SMO_DmrC_T2098L_CeGFP	SMO target fused to DmrC for chemically induced recruitment. CeGFP enables fluorescence-based detection.
MGRN1_DmrA_His6	Recruits MGRN1 to SMO to test whether induced proximity increases SMO ubiquitination.
MGRN1_C279A_C282A_DmrA_His6	Catalytic-site mutant control to assess the dependence of ubiquitination on MGRN1 activity.
MEGF8_ICD_2602_2778_DmrA_His6	Recruits the MEGF8 intracellular domain to SMO to investigate whether endogenous MGRN1 can mediate ubiquitination.
DmrA_mCherry_proximity_only_control	Negative control for recruitment of an unrelated protein without MGRN1 ligase activity.
GPR161_EGFP_DmrC_T2098L	Additional ciliary receptor target to investigate whether the recruitment strategy can be extended beyond SMO.

ALFA/NbALFA recruitment system (no chemical inducer)

Construct	Purpose
SMO_His6_ALFA_v2	ALFA-tagged SMO target for NbALFA-mediated recruitment.
HA_MGRN1_NbALFA	MGRN1 fused to an ALFA-binding nanobody to facilitate recruitment to SMO.
HA_MGRN1_C279A_C282A_NbALFA	Catalytic-site mutant control to evaluate whether the response depends on MGRN1 activity.
GPR161_GFP_ALFA	Additional ALFA-tagged ciliary receptor to assess applicability beyond SMO.

The C279A/C282A constructs are designed as catalytic-activity controls. Their loss of ligase activity should be verified experimentally rather than assumed.

Materials and equipment

- NIH/3T3 cells and growth medium
- The ten GenScript constructs above, purified by midiprep
- AP21967 (Takara A/C Heterodimerizer) and vehicle
- Transfection reagent PEI
- ALFA Selector resin (NanoTag)
- Antibodies: SMO, ubiquitin, HA, His6, MGRN1, loading control GAPDH
- Reagents for lysis, SDS-PAGE, Western blotting and BCA protein assay
- Chemiluminescence imager

Experimental conditions

Condition	SMO construct	Recruited protein	AP21967
Direct MGRN1 recruitment	SMO_DmrC_T2098L_CeGFP	MGRN1_DmrA_His6	Vehicle and AP21967
Catalytic-site mutant	SMO_DmrC_T2098L_CeGFP	MGRN1_C279A_C282A_DmrA_His6	AP21967
MEGF8-mediated recruitment	SMO_DmrC_T2098L_CeGFP	MEGF8_ICD_2602_2778_DmrA_His6	Vehicle and AP21967
Proximity-only control	SMO_DmrC_T2098L_CeGFP	DmrA_mCherry_proximity_only_control	AP21967

SMO alone and dimerizer only	SMO_DmrC_T2098L_CeGFP	None	Vehicle and AP21967
ALFA/NbALFA recruitment	SMO_His6_ALFA_v2	HA_MGRN1_NbALFA	None
ALFA/NbALFA catalytic-site mutant	SMO_His6_ALFA_v2	HA_MGRN1_C279A_C282A_NbALFA	None
GPR161, DmrA/DmrC	GPR161_EGFP_DmrC_T2098L	MGRN1_DmrA_His6	Vehicle and AP21967
GPR161, ALFA/NbALFA	GPR161_GFP_ALFA	HA_MGRN1_NbALFA	None

Procedure: plasmid preparation

1. Amplify each construct: transform into DH5α, grow overnight at 37 °C in 50 mL LB with carbenicillin (1:1,000 dilution of stock).
2. Purify each construct by midiprep (Experiment 2, Part B).

Procedure: seeding and transfection

1. Seed NIH/3T3 cells in 6-well plates at 2.5×10^5 cells per well in 2 mL of DMEM (4.5 g/L glucose, L-glutamine) with 10% calf serum, 24 hours before transfection. Cells should be about 70% confluent at transfection.
2. Transfect each condition with its plasmid combination from the table above using PEI-MAX. Use 1.9 µg of each plasmid per well, which is 3.75 µg of total DNA for a two-plasmid condition. Keep total DNA the same in every well by topping up conditions with fewer plasmids
3. In a sterile hood, make a 1:10 dilution of PEI-MAX (1 µg/µL stock) in sterile 150 mM NaCl and mix thoroughly by inversion. For each well, combine 37.5 µL of the diluted PEI with 171 µL of sterile 150 mM NaCl.
4. Add the PEI mixture to the DNA and mix by pipetting up and down 30 times.
5. Incubate at room temperature for 12–15 minutes.
6. Add the mixture to the cells very slowly, almost dropwise, without touching the liquid in the well. Tilt the plate toward you and hold the pipette away from you. Return the plate to the 37 °C incubator.

Procedure: AP21967 treatment

1. Dilute AP21967 in culture medium to the final concentration of 500 nM, the top of the manufacturer's suggested range of 0.05–500 nM (a 1:1,000 dilution of a 0.5 mM stock). Add the same volume of vehicle ethanol to control wells.
2. Add to the cells 24 hours after transfection and incubate for 6 hours. ALFA/NbALFA conditions receive no inducer and are harvested at the same time.

3. Harvest the cells for SMO abundance, co-immunoprecipitation and the ubiquitination assay by seeding and transfecting three wells per condition, one for each assay, because the lysis buffers differ. Rinse each well with cold PBS and lyse it directly in the buffer for its assay.

Procedure: SMO abundance (total lysate Western blot)

1. Lyse cells in RIPA lysis buffer (50 mM HEPES pH 7.9, 140 mM NaCl, 1 mM EDTA, 1% Triton X-100, 0.1% SDS, 0.1% sodium deoxycholate) with protease and phosphatase inhibitors, for 30 minutes at 4 °C on a rotator.
2. Clear the cell debris by centrifugation at 2000 xg, then measure protein concentration by BCA.
3. Load 10 µL of protein per lane onto a Bis-Tris PAGE gel, run the gel at standard conditions, and transfer to a PVDF membrane.
4. Block with 5% milk in TBS-T (20 mM Tris HCl pH 7.5, 150 mM NaCl, 0.1% Tween 20) for 1 hour.
5. Incubate with primary antibodies at 4 °C overnight. Wash with TBS-T, then incubate with an HRP secondary antibody for 1 hour at room temperature.
6. Visualise with a chemiluminescent HRP substrate on the imager.
7. Quantify SMO relative to a loading control GAPDH and to the vehicle condition.

Procedure: co-immunoprecipitation

1. Lyse cells in a non-denaturing lysis buffer with protease inhibitors and clear by centrifugation. Keep a sample of the lysate as input.
2. Capture the bait on affinity resin. For ALFA-tagged SMO, use ALFA Selector resin following the NanoTag immunoprecipitation manual. For SMO_DmrC_T2098L_CeGFP, capture with a GFP-binding resin.
3. Wash the resin twice and elute in SDS sample buffer.
4. Run Western blots of input and eluate. Probe for the recruited protein (anti-His for the DmrA constructs, anti-HA for HA_MGRN1_NbALFA) and, in the MEGF8 condition, for endogenous MGRN1 with MGRN1 antibody.

Procedure: ubiquitination assay (denaturing)

1. Rinse cells with cold PBS and lyse in buffer containing 2% SDS and a deubiquitinase inhibitor such as N-ethylmaleimide.
2. Boil for 10 minutes and sonicate until the lysate is clear.
3. Dilute the lysate about tenfold in lysis buffer without SDS, so the SDS concentration falls to about 0.2%.
4. Capture SMO from the diluted lysate: for SMO_His6_ALFA_v2, with ALFA Selector resin, which tolerates denaturing conditions; for SMO_DmrC_T2098L_CeGFP, with a GFP-binding resin. Wash in buffer containing 0.1% SDS.
5. Elute in SDS sample buffer and run a Western blot. Probe for ubiquitin and for SMO. Ubiquitin is detected from endogenous ubiquitin; no ubiquitin construct is among the ten.

Controls

- Vehicle-treated cells for each DmrA/DmrC condition.
- Catalytically inactive MGRN1 (C279A/C282A) in both recruitment systems.
- DmrA_mCherry proximity-only control with AP21967.
- ALFA/NbALFA conditions with no inducer.

Expected readout and analysis

- Co-immunoprecipitation: AP21967 is expected to increase association between proteins carrying DmrA and DmrC. Co-immunoprecipitated amounts are normalised to the recovered tagged protein, which accounts for differences in expression or recovery rather than in recruitment.
- Ubiquitination: more ubiquitin signal on isolated SMO with directly recruited MGRN1 than in the controls would show that bringing the ligase to SMO is sufficient. Pulling down endogenous MGRN1 together with recruited MEGF8, with more ubiquitination, would support recruitment of the machinery the cell already has.
- SMO abundance: SMO is expected to decrease when recruitment promotes its degradation.
- A decrease in SMO abundance is read alongside the ubiquitination result. On its own it does not show which degradation pathway is involved.

References

- Kong et al., 2020. Developmental Cell. https://pmc.ncbi.nlm.nih.gov/articles/PMC7686252/
- Takara Bio, A/C Heterodimerizer (identical to AP21967), recommended concentration range: https://takarabio.com/x9691.xml
- Takara Bio, A/C Heterodimerizer certificate of analysis (0.5 mM in 100% ethanol, catalog 635057): https://www.takarabio.com/documents/Certificate%20of%20Analysis/635057/635057-013117.pdf
- NanoTag, ALFA Selector immunoprecipitation protocol: https://sysy.com/products/protocols/ALFA_Selector_Manual_1.pdf
- Detection of Protein Ubiquitination, JoVE: https://jove.com/t/1293
- Ubiquitination assay under denaturing conditions, Bio-protocol: https://bio-protocol.org/prep1594